

# DengAI: Predicting Dengue Fever Cases Using Machine Learning

BLG527E – Machine Learning Term Project Fall 2025

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itü



# Problem & Goal

## Problem

- Dengue fever is a mosquito-borne disease
- Strongly affected by climate & environmental factors

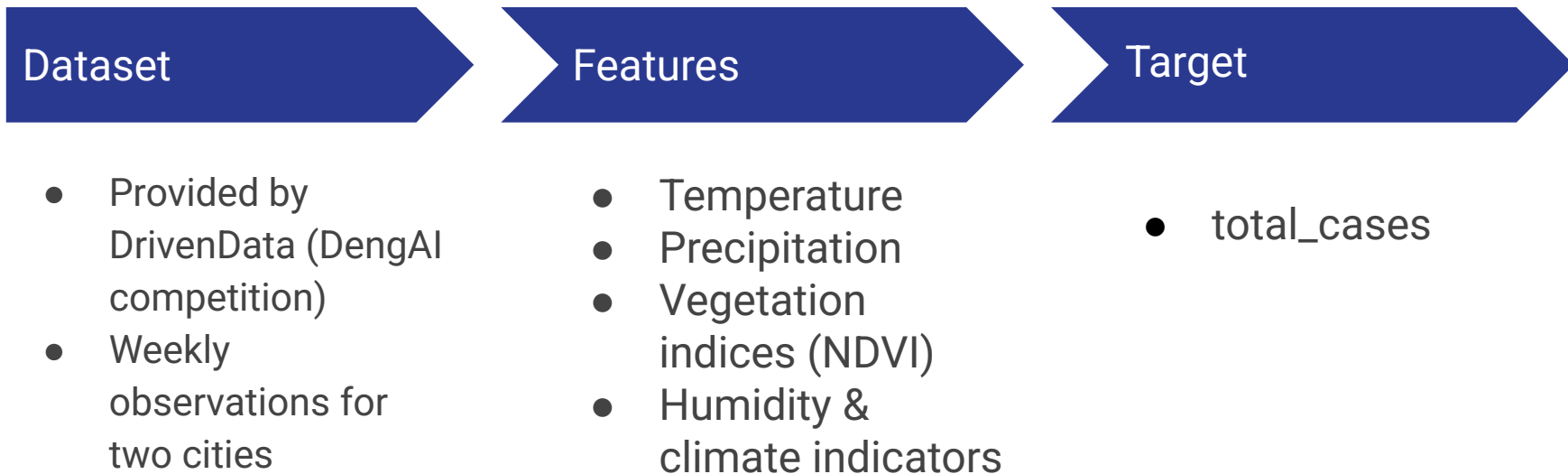
## Goal

- Predict weekly dengue case counts
- For two cities: San Juan (Puerto Rico) Iquitos (Peru)

## Evaluation Metric

- Mean Absolute Error (MAE)

# Dataset & Features



## Key observation:

San Juan → higher variance, stronger outbreaks | Iquitos → lower, more stable case counts

# Data Preprocessing

## Missing values:

- Imputed with city-specific means

## City-aware modeling:

- San Juan and Iquitos processed separately
- Time-aware split: **80% train / 20% validation**
- No shuffling (prevents leakage)

# Feature Engineering

Seasonal encoding: **weekofyear**

- Climate & vegetation features

Time-series structure:

- Weekly ordering preserved
- Separate models per city
- Enables fair temporal evaluation

# Models Used

We evaluated two main approaches:

1. **Polynomial Regression (Baseline)**
  2. **Literature-Based Model Gradient Boosting Regression**
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# Model 1: Polynomial Regression

Baseline nonlinear regression

- Polynomial feature expansion:
- Degrees tested:  $k = 2, 3, 4$
- Ordinary Least Squares regression
- Evaluated using MAE on time-based split

Purpose:

- Simple baseline
  - Reference for advanced models
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**Table 2: Validation MAE of polynomial regression for different polynomial degrees  $k$  using weekly prediction.**

| City     | $k = 2$ | $k = 3$ | $k = 4$ |
|----------|---------|---------|---------|
| San Juan | 53.52   | 303.77  | 646.93  |
| Iquitos  | 15.41   | 46.05   | 55.96   |

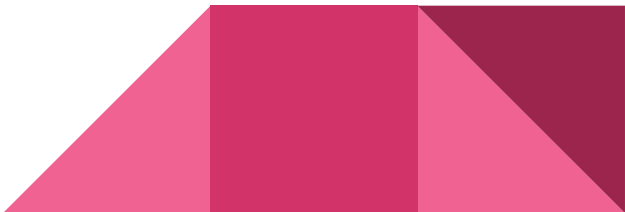
# Model 2: Gradient Boosting Regression

- Ensemble of shallow regression trees

Advantages:

- Captures nonlinear relationships
- Handles feature interactions

Hyperparameters:

- 300 estimators
  - Learning rate = 0.05
  - Max depth = 3
  - Trained separately per city
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# Weekly Prediction Results

**Polynomial Regression (MAE):** Performance worsens with higher degree  
Strong overfitting, especially for San Juan

**Gradient Boosting (MAE):** Substantially outperforms polynomial model  
Especially effective for volatile outbreaks

| City          | Polynomial ( $k = 2$ ) | Gradient Boosting |
|---------------|------------------------|-------------------|
| San Juan (sj) | 53.52                  | <b>18.31</b>      |
| Iquitos (iq)  | 15.41                  | <b>7.51</b>       |

# Feature Importance

**Table 4: Top 10 features ranked by Gradient Boosting importance for San Juan.**

| Feature                               | Importance |
|---------------------------------------|------------|
| year                                  | 0.524      |
| weekofyear                            | 0.175      |
| reanalysis_specific_humidity_g_per_kg | 0.068      |
| reanalysis_dew_point_temp_k           | 0.047      |
| station_max_temp_c                    | 0.030      |
| ndvi_se                               | 0.024      |
| reanalysis_max_air_temp_k             | 0.020      |
| station_precip_mm                     | 0.019      |
| reanalysis_avg_temp_k                 | 0.016      |
| reanalysis_precip_amt_kg_per_m2       | 0.013      |

**Table 5: Top 10 features ranked by Gradient Boosting importance for Iquitos.**

| Feature                               | Importance |
|---------------------------------------|------------|
| year                                  | 0.240      |
| weekofyear                            | 0.190      |
| reanalysis_specific_humidity_g_per_kg | 0.102      |
| reanalysis_avg_temp_k                 | 0.060      |
| reanalysis_tdtr_k                     | 0.056      |
| ndvi_sw                               | 0.054      |
| reanalysis_dew_point_temp_k           | 0.047      |
| reanalysis_relative_humidity_percent  | 0.037      |
| reanalysis_precip_amt_kg_per_m2       | 0.032      |
| ndvi_se                               | 0.030      |

# Forecasting Horizon Results

**Table 6: Validation MAE of Gradient Boosting Regression for early prediction at different forecasting horizons  $h$  (in weeks). Lower values indicate better performance.**

| City          | $h = 1$ | $h = 2$ | $h = 3$ | $h = 4$ |
|---------------|---------|---------|---------|---------|
| San Juan (sj) | 17.54   | 17.82   | 18.25   | 18.72   |
| Iquitos (iq)  | 7.57    | 9.15    | 8.06    | 11.17   |

# Results & Performance



#1622 out of 15,910

DengAI: Predicting Disease Spread

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Best score

25.4014

Current rank

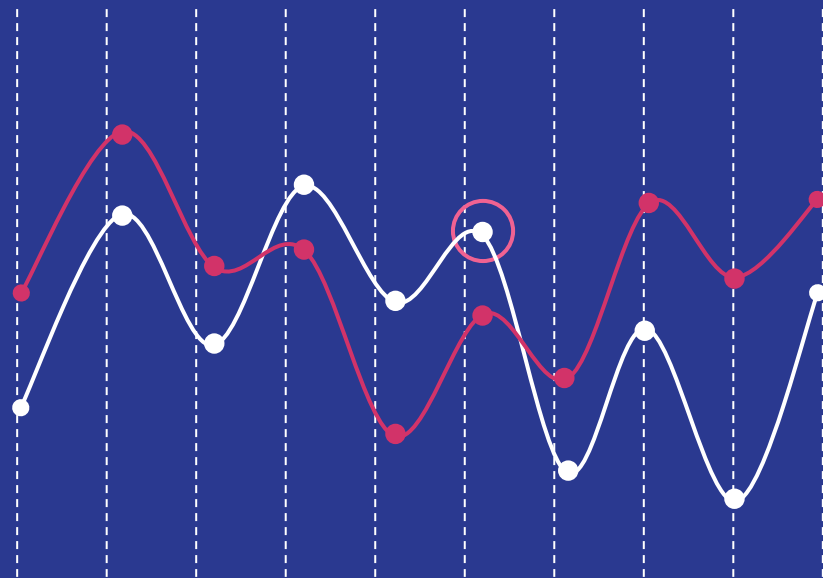
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Submissions used

1 of 3

Make new submission

You have **2 of 3** submissions left today. Your next submission can be on Jan. 14, 2026 UTC.

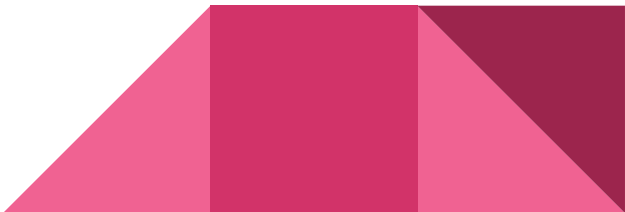


# Conclusion & Future Work

## Conclusion

- Machine learning can effectively predict dengue outbreaks
- Random Forest outperformed polynomial regression
- Best forecasting horizon: 1 weeks ahead

## Future Work

- Use ensemble models (XGBoost, stacking)
  - Incorporate spatial data
  - Explore deep learning for time series
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# References

- [1] Arunans23. 2019. DengAI-datadriven: Dengue Fever Prediction Using Machine Learning. <https://github.com/arunans23/DengAI-datadriven>. Accessed: January 14, 2026.
- [2] DrivenData. 2016. DengAI: Predicting Disease Spread. <https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/>. Accessed: January 14, 2026.
- [3] Z. J. Madewell, D. M. Rodriguez, M. B. Thayer, et al. 2025. Machine learning for predicting severe dengue in Puerto Rico. *Infectious Diseases of Poverty* 14, 5 (2025). doi:10.1186/s40249-025-01273-0
- [4] Md. Siddikur Rahman and Md. Abu Bokkor Shiddik. 2025. Explainable artificial intelligence for predicting dengue outbreaks in Bangladesh using eco-climatic triggers. *Global Epidemiology* 10 (2025). doi:10.1016/j.gloepi.2025.100210

